DATE:

EX: 4 COMPUTE ACF FOR THE FIRST 35 LAGS

AIM

To compute and plot the AutoCorrelation Function (ACF) of a time-series dataset for the first 35 lags and use the ACF pattern to identify a suitable time-series model.

ALGORITHM

1. Start the program.
2. Import Pandas, Matplotlib, acf, and plot_acf from statsmodels.
3. Create the given time-series data and convert it into a Pandas Series.
4. Display the time-series values.
5. Plot the original time-series data.
6. Calculate the AutoCorrelation Function for the first 35 lags using nlags = 35.
7. Display the ACF value for each lag from 0 to 35.
8. Plot the ACF graph for the first 35 lags.
9. Observe the ACF pattern to identify whether AR, MA, or ARMA may be suitable.
10. Stop the program.

PROGRAM

```
import pandas as pd
import matplotlib.pyplot as plt
from statsmodels.tsa.stattools import acf
from statsmodels.graphics.tsaplots import plot_acf

data = [
    112, 118, 132, 129, 121, 135,
    148, 148, 136, 119, 104, 118,
    115, 126, 141, 135, 125, 149,
    170, 170, 158, 133, 114, 140,
    145, 150, 165, 160, 155, 170,
    190, 188, 175, 160, 145, 165,
    170, 180, 195, 190, 185, 200
]

series = pd.Series(data)

print("Time Series Data:")
print(series)

plt.figure(figsize=(10, 4))
plt.plot(series, marker='o')
plt.title("Time Series Data")
plt.xlabel("Time")
plt.ylabel("Value")
plt.grid(True)
plt.show()

acf_values = acf(series, nlags=35)

print("\nACF Values:")
for lag, value in enumerate(acf_values):
    print("Lag", lag, ":", round(value, 4))

plt.figure(figsize=(12, 5))
plot_acf(series, lags=35)
plt.title("ACF for First 35 Lags")
plt.xlabel("Lag")
plt.ylabel("Autocorrelation")
plt.show()
```

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OUTPUT

Time-Series Values

```
0    112
1    118
2    132
3    129
4    121
5    135
6    148
7    148
8    136
9    119
10   104
11   118
12   115
13   126
14   141
15   135
16   125
17   149
18   170
19   170
20   158
21   133
22   114
23   140
24   145
25   150
26   165
27   160
28   155
29   170
30   190
31   188
32   175
33   160
34   145
35   165
36   170
37   180
38   195
39   190
40   185
41   200
dtype: int64
```

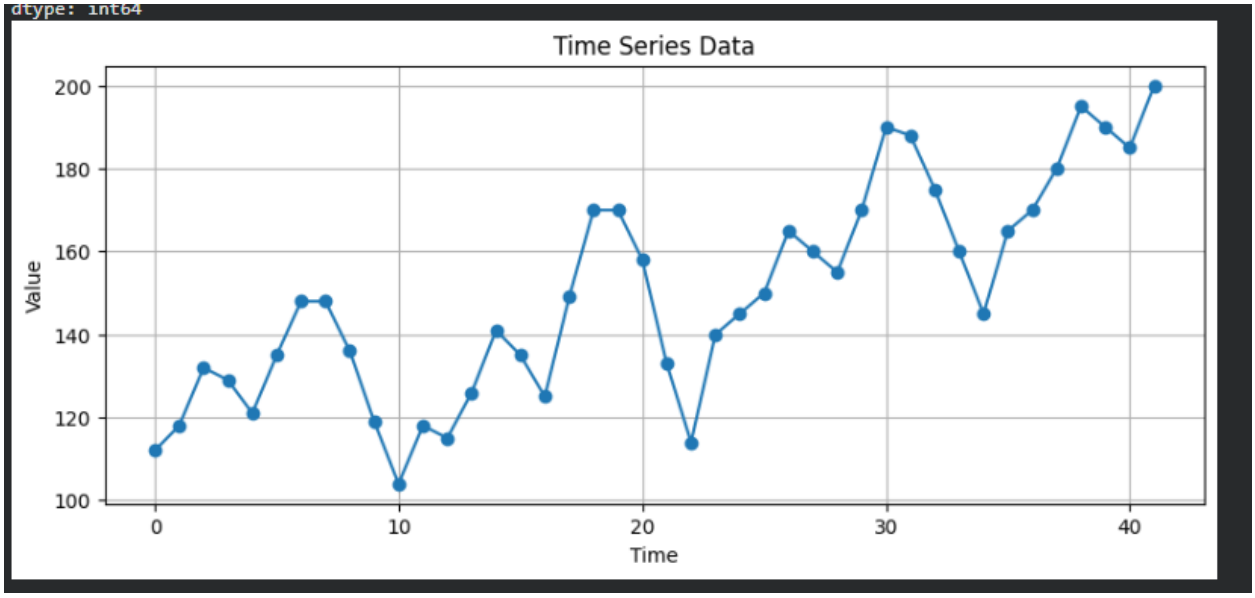
ACF Values

```
ACF Values:
Lag 0 : 1.0
Lag 1 : 0.7916
Lag 2 : 0.5577
Lag 3 : 0.4112
Lag 4 : 0.3091
Lag 5 : 0.2953
Lag 6 : 0.318
Lag 7 : 0.2762
Lag 8 : 0.2813
Lag 9 : 0.3086
Lag 10 : 0.3284
Lag 11 : 0.351
Lag 12 : 0.3334
Lag 13 : 0.1568
Lag 14 : -0.0186
Lag 15 : -0.1155
Lag 16 : -0.1906
Lag 17 : -0.2068
Lag 18 : -0.1844
Lag 19 : -0.1845
Lag 20 : -0.1191
Lag 21 : -0.066
Lag 22 : -0.0858
Lag 23 : -0.1407
Lag 24 : -0.1884
Lag 25 : -0.2799
Lag 26 : -0.3316
Lag 27 : -0.3496
Lag 28 : -0.3738
Lag 29 : -0.3674
Lag 30 : -0.3159
Lag 31 : -0.2608
Lag 32 : -0.1724
Lag 33 : -0.1199
Lag 34 : -0.1217
Lag 35 : -0.1628
<Figure size 1200x500 with 0 Axes>
```

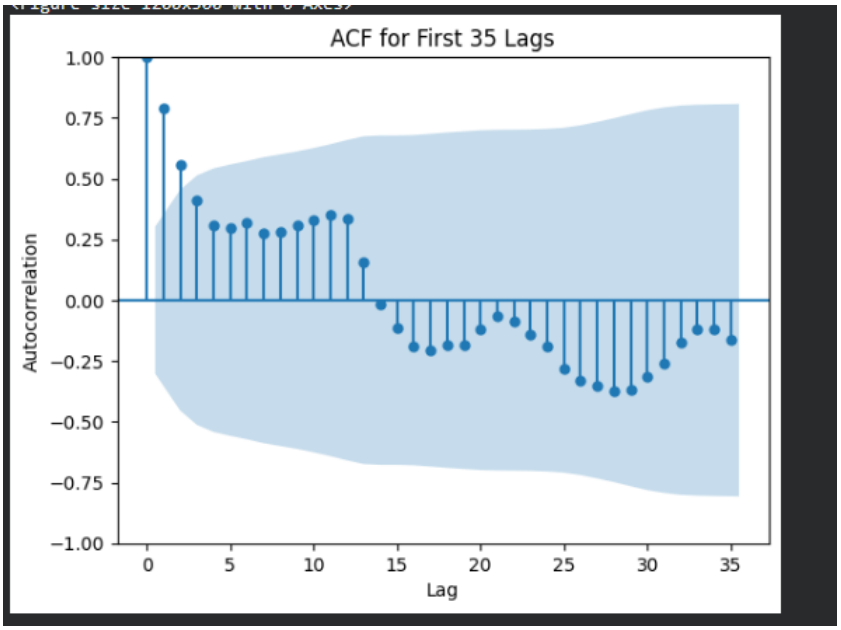
DATE:

GRAPHS

Time Series Data



ACF for First 35 Lags



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OBSERVATION

The ACF starts with a high positive value at the first lag and then gradually decreases. At higher lags, the autocorrelation becomes negative and later moves back toward zero. The pattern tails off rather than cutting off sharply, so an AR-type or ARMA-type structure may be considered. ACF alone is not sufficient to determine the exact model order; PACF should also be examined.

RESULT

Thus, the ACF of the given time-series data was successfully calculated and plotted for the first 35 lags, and the ACF pattern was examined for possible time-series model identification.